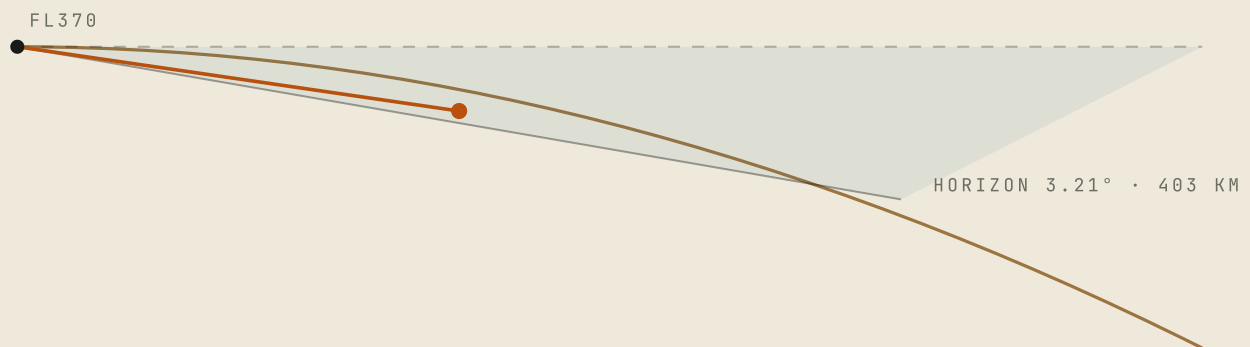


NADIR N. THE POINT ON THE GROUND DIRECTLY BENEATH AN OBSERVER

What *is* that?

Aim a phone out of an aircraft window and it names the ground below.
With the radio off, because the entire atlas is already on the device. And it
shows you how wrong it might be, instead of a pin it cannot justify.



SECTION THROUGH THE LINE OF SIGHT · VERTICAL EXAGGERATION

HORIZON AT CRUISE

3.21°

below level, not thirty

NADIR · CASE STUDY

AND IT IS

403 km

away, which is the problem

ATLAS ON DEVICE

1.09 MB

13,394 named features

Everyone asks. Nobody answers.

There is a question every window seat produces and no product has ever answered well: what is that. A river bending through a plain, a coastline, a city at night. You look at it for ninety seconds and then it is gone and you never find out.

The obvious answer is a map app, and the obvious answer fails for two reasons. There is no network at cruise, and a top-down map does not answer the question you actually asked. You did not ask where you are. You asked what *that* is, out of the window, in a direction, at a distance.

Flyover Country, which is NSF funded and free, is the closest existing product and it is genuinely good. It caches your flight path and shows you geology offline. But it is a map: you look down at a dot. Nobody has built the thing where you point the camera and it tells you.

Three constraints that decide everything

No network. Not slow, none. Anything fetched at 11 km is not going to arrive.

No trustworthy compass. A magnetometer inside an aluminium tube is a rumour, off by fifteen degrees or worse.

Brutal angular sensitivity. At a 300 km slant range, one degree of error is five kilometres of ground.

Any one of those makes the naive build wrong. Together they mean the product is not a mapping problem at all. It is a geometry problem with an error budget, wearing a mapping problem's clothes.

VERIFIED, NOT ASSUMED

GPS works in aeroplane mode on both iOS and Android. The receiver is passive: it only listens to satellites, so switching off the cellular, wifi and Bluetooth transmitters does not touch it. That single fact is what makes the product possible on both platforms rather than Android only, and it was checked rather than assumed.

Three degrees of everything

At cruise the horizon sits 3.21 degrees below level. The entire visible world, out to 403 kilometres, is folded into a band of sky thinner than a thumb at arm's length.

The observer sits at a known position and altitude, looking along a known bearing at a known angle below the horizontal. Where does that line of sight meet the ground? A flat earth answers $d = h / \tan(\text{dep})$. That answer is fine for a drone and useless for an airliner, because as the look flattens towards the horizon it runs off to infinity while the real one converges on the limb.

Solved on a sphere instead. O is the earth's centre, A the observer at radius $r = R + h$, P the ground intersection at radius R:

```
angle OAP = 90 - dep           (ray off the downward vertical)
sin(OPA)  = (r / R) * cos(dep) (law of sines)

gamma     = asin((r/R)*cos(dep)) + dep - 90

null when (r/R)*cos(dep) > 1    (the ray clears the limb)
```

Returning null there is the difference between saying "sky" and inventing a ground point, and it is why the near-horizon behaviour tells the truth.

What the flat approximation costs

LOOK DOWN	FLAT EARTH	SPHERE	FLAT IS SHORT BY
45°	11.3 km	11.3 km	0.1%
20°	31.0 km	31.2 km	0.6%
10°	64.0 km	65.7 km	2.6%
5°	128.9 km	145.8 km	11.6%
4°	161.3 km	201.7 km	20.0%
3.21°	201 km	403 km	the horizon

The surface curves away beneath the ray, so the ground has dropped by the time the ray arrives. At the horizon the flat model is out by a factor of two.

And the view does not recede evenly

10 km of ground sits 48.51° below level. 100 km sits at 6.84°. 400 km sits at 3.21°. The first ten kilometres take up more of your window than everything between one hundred and four hundred put together.

Do not trust the compass

The tempting build reads the magnetometer, because it hands you a bearing for free. Inside a fuselage, beside a seat-back screen, a power bus and three hundred other phones, that bearing is worth nothing. Fifteen degrees of error is normal. Fifteen degrees at a 300 km slant range is a hundred kilometres of ground.

So heading comes from somewhere else entirely. Differentiate consecutive GPS fixes and the aircraft's course over ground falls out to a fraction of a degree, with no magnetic interference at all. An airliner in cruise flies a very straight line very fast, which is the ideal case for that measurement.

The window is the calibration

That gives the aircraft's heading, not the phone's. Turning one into the other needs a single known relationship, and the cabin supplies it free: a window is perpendicular to the direction of flight. Hold the phone flat against the glass, say which side you are sitting on, and the offset is solved without the magnetometer being consulted once.

Depression is the easy half. It comes from the gravity vector, which is absolute and needs no calibration. The one thing that breaks it is a turn: in a coordinated turn the accelerometer reads perpendicular to the wings rather than towards the centre of the earth, so down stops pointing down. The app detects the turn from the GPS track and declines to answer until the wings are level.

MAGNETOMETER, IN A CABIN

± 15°

which is 100 km of ground at cruise range

HEADING FROM THE GPS TRACK

± 0.5°

a thirtyfold improvement, for free, from data already on hand

The error budget

SOURCE	1σ
GPS position	10 m
GPS altitude	30 m
Heading, track derived	0.5°
Pitch, still air	1.0°
Pitch, turbulence	3.0°

Position barely matters and angle is everything. It is an angle problem wearing a location problem's clothes.

A pin would be a *lie*

Every other product would drop a marker and let you assume it was right. The geometry does not support that. What it supports is an ellipse, and the ellipse is a very strange shape.

Propagating the sensor errors through the ray-cast gives a ground-space ellipse. Across the line of sight it stays sharp, because track-derived heading is good to half a degree. Along the line of sight it smears, because ground distance is savagely sensitive to look angle near the horizon. **The two axes are perpendicular and they never mix.** That is why the shape is a sliver and never a circle, and why the app can tell you which river you are over while having no idea how far along it you are.

LOOK DOWN	LANDS AT	ERROR ALONG	ERROR ACROSS	VERDICT
80°	2.0 km	0.2 km	0.02 km	Pinned
45°	11.3 km	0.4 km	0.10 km	Pinned
15°	42.6 km	3.0 km	0.37 km	Pinned
10°	65.8 km	7.1 km	0.57 km	Confident
5°	146.1 km	38.3 km	1.28 km	Approximate
4°	202.2 km	84.8 km	1.76 km	Coarse
3.3°	317.4 km	unbounded	2.77 km	Unresolved

The measurement the product is built on

0.001° NEAR THE HORIZON

9.9 km

of ground

THE IDENTICAL 0.001° AT 60° DOWN

0.3 m

of ground

The same angular twitch is worth thirty-three thousand times more ground near the limb, because ground distance goes as the square root of the angle below the dip. This is not a claim, it is a measurement, and it is printed by the test suite.

Which makes the resolver's job a different question

Not "what is nearest" but "what is the most specific thing I am entitled to say, given how wrong I might be". A feature only qualifies if it is at least as large as the error radius. Below that, choosing it over its neighbours is a guess in a confident font.

on the Krishna ±1 km → Krishna (river) ±5 → Eastern Ghats ±250 → India
over Delhi ±1 km → New Delhi ±20 → Delhi ±80 → Ganges Plain
Everest ±1 km → Mount Everest ±80 → Himalayas
Vizag ±1 km → Vishakhapatnam ±20 → Bay of Bengal

The whole world, 1.09 MB

Nothing is fetched in flight, so everything the resolver can name has to already be on the device. Three primitives come out of that, because three different kinds of question get asked of a point on the ground: points match by radius, polygons by containment, lines by distance.

The source is Natural Earth, public domain, simplified with an iterative Douglas-Peucker and rounded to about a hundred metres. Nothing on board needs more precision, because the sensors are the limit long before the map is.

Because it is all local there is no account, no telemetry and nothing to leak. The position never leaves the phone for the simple reason that there is nowhere for it to go.

Honest limits

- State boundaries currently cover nine countries, which happen to be the large flight geographies.
- The gazetteer is Natural Earth rather than a full GeoNames extract, so rural areas fall back to regions sooner than they need to.
- Both are pack upgrades, not engine changes.

LAYER	COUNT
Towns and cities	7,342
River segments	2,289
Coastline runs	1,453
Named summits	644
Lakes	313
States and provinces	292
Islands	267
Countries	215
Mountain ranges	129
Seas, bays, gulfs	114
Capes	111
Plateaus	45
Deserts	35
Deltas	12

COMPRESSED

1.09 MB

FEATURES

13,394

REQUESTS MADE IN FLIGHT

0

Drawing the atlas in perspective

The demo has to be honest on a laptop, so it runs two ways into one engine. On a phone above ten kilometres the sensors are real. Anywhere else a scripted flight supplies the two things a laptop genuinely cannot, a position at altitude and a direction of flight. Everything downstream is the same code, and both emit the identical observer shape, so the replay is a demonstration rather than a mock-up.

The main view is not a map. It is the view out of the window, reconstructed from vectors: every river, coastline and town projected through a real pinhole camera model from the aircraft's position, using the closed-form inverse of the ray-cast.

```
depression = atan2(r - R*cos(gamma), R*sin(gamma))
```

90° at the nadir, exactly the dip angle at the horizon,
no iteration. Round-trips against castRay to 2.4e-12 degrees.

Which makes the argument visible in a way prose cannot. Range arcs drawn at 25, 50, 100, 200 and 400 km stack up ever closer together towards the horizon. You can see the ground being crushed. The picture and the maths are the same argument.

Art direction

A VFR sectional chart, not an illustration of one: buff stock rather than white, cyan for water, tan for terrain relief, hairline neatlines, tiny mono annotation on everything. Type is either a serif making an argument or mono stating a fact, with nothing in between.

The single action colour is aviation amber. It is what the industry has always used to mean "caution, pay attention to this", which for a product whose entire subject is how wrong it might be, is the correct accent.

Where the honesty shows up in the design

- The answer is never a pin. It is an ellipse lying along the line of sight.
- The result card never covers the frame it is identifying.
- As confidence drains, the typography loses confidence with it: the answer goes italic and fades.
- Past the limb it says "sky" rather than guessing.
- Every number on screen is computed live, not written into the copy.

Five dollars, once

There is no server, so there is nothing to charge rent for. No subscription, no account, no advertising, no analytics. A subscription here would be indefensible and travellers punish it.

The precedent is exact. SkyView is a one-time purchase of about five dollars and has sold in the millions doing this for the sky. NADIR is SkyView pointed down.

Web is the demo, native is the product

The web build proves the engine and is free forever. The app is what you want in seat 21A: the barometer for altitude instead of GPS, proper sensor fusion rather than DeviceOrientation, and packs that are still on the device a fortnight after download.

Open the engine, sell the app

The geometry and uncertainty modules are MIT. It is what makes "we tell you when we do not know" verifiable rather than a marketing line, and it gives up nothing: the product is the app, the packs and the sensor work.

What is built

- Spherical ray-cast with refraction, and a test suite that passes
- Error propagation to a ground ellipse with confidence bands
- Resolver with graceful degradation from landmark to region
- 13,394 feature offline pack, built from public-domain data
- Perspective window view and track-up chart, both from the pack
- Live sensor path and scripted replay, one engine
- Service worker: the page itself works offline

What is next

- Native build, for the barometer and durable storage
- GeoNames gazetteer for rural density
- Per-route pack slicing, so you carry a corridor not a planet
- Real device testing at altitude, which no amount of maths substitutes for

DATA

Natural Earth, public domain. Coastlines, rivers, physical regions, administrative boundaries and populated places.

You will still not know what most of it is called. But you will know when you *do*.